ASSIGNMENT 10.3

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TASK 1 :

task1.py | task2.py | task3.py | task4.py | task5.py | task6.py

task1.py > ...

```python
def print_rectangle_details(length, width):
    None
    """
    area = length * width
    perimeter = 2 * (length + width)

    print("Area of Rectangle:", area)
    print("Perimeter of Rectangle:", perimeter)


# Rectangle 1
print_rectangle_details(5, 10)

# Rectangle 2
print_rectangle_details(7, 12)

# Rectangle 3
print_rectangle_details(10, 15)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task1.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task1.py"
Area of Rectangle: 50
Perimeter of Rectangle: 30
Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 2 :

```python
#Refactoring – Extracting Reusable Functions
def calculate_total(price):
    """
    Calculate total price including 18% tax.

    Parameters:
    price (float): Base price

    Returns:
    float: Total price including tax
    """
    tax = price * 0.18
    return price + tax


prices = [250, 500]

for p in prices:
    print("Total Price:", calculate_total(p))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task2.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task2.py"
Total Price: 295.0
Total Price: 590.0
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 3 :

task1.py | task2.py | task3.py | task4.py | task5.py | task6.py

task3.py > ...

```python
#Refactoring Using Classes and Methods
class GradeCalculator:
    """
    A class to calculate student grades based on marks.
    """

    def calculate_grade(self, marks):
        """
        Determine grade based on marks.

        Parameters:
        marks (int): Student marks (0-100)

        Returns:
        str: Grade result
        """
        if 90 <= marks <= 100:
            return "Grade A"
        elif marks >= 80:
            return "Grade B"
        elif marks >= 70:
            return "Grade C"
        elif marks >= 40:
            return "Grade D"
        else:
            return "Fail"


calculator = GradeCalculator()

marks_list = [85, 72, 95, 35]

for m in marks_list:
    print(f"Marks: {m} → {calculator.calculate_grade(m)}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task3.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task3.py"
Marks: 85 → Grade B
Marks: 72 → Grade C
Marks: 95 → Grade A
Marks: 35 → Fail
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

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TASK 4 :

task4.py > ...

```python
#Refactoring – Converting Procedural Code to Functions
def calculate_area(radius):
    """
    Calculate the area of a circle.

    Parameters:
    radius (float): Radius of the circle

    Returns:
    float: Area of the circle
    """
    import math
    return math.pi * radius ** 2
radii = [3, 5, 7]
for r in radii:
    print("Area of Circle:", calculate_area(r))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task4.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task4.py"
Area of Circle: 28.274333882308138
Area of Circle: 78.53981633974483
Area of Circle: 153.93804002589985
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 5 :

task4.py > ...

```python
#Refactoring – Converting Procedural Code to Functions
def calculate_area(radius):
    """
    Calculate the area of a circle.

    Parameters:
    radius (float): Radius of the circle

    Returns:
    float: Area of the circle
    """
    import math
    return math.pi * radius ** 2
radii = [3, 5, 7]
for r in radii:
    print("Area of Circle:", calculate_area(r))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task4.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task4.py"
Area of Circle: 28.274333882308138
Area of Circle: 78.53981633974483
Area of Circle: 153.93804002589985
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 6 :

task6.py > ...

```python
def check_access(users_set, username):
    users_set (set): A set containing valid usernames.
    username (str): The username to search for.

    Returns:
    bool: True if user exists, False otherwise.
    """
    return username in users_set


# Using a set for O(1) lookup
users = {"admin", "guest", "editor", "viewer"}

name = input("Enter username: ")

if check_access(users, name):
    print("Access Granted")
else:
    print("Access Denied")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task6.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task6.py"
Enter username: admin
Access Granted
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 7 :

task6.py > ...

```python
def check_access(users_set, username):
    users_set (set): A set containing valid usernames.
    username (str): The username to search for.

    Returns:
    bool: True if user exists, False otherwise.
    """
    return username in users_set


# Using a set for O(1) lookup
users = {"admin", "guest", "editor", "viewer"}

name = input("Enter username: ")

if check_access(users, name):
    print("Access Granted")
else:
    print("Access Denied")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task6.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task6.py"
Enter username: admin
Access Granted
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 8 :

task8.py > ...

```python
#task 8
#Fibonacci Generator
def fibonacci(n):
    """
    Generate Fibonacci sequence up to n terms.

    Parameters:
    n (int): Number of terms to generate

    Returns:
    list: Fibonacci sequence up to n terms
    """
    sequence = []
    a, b = 0, 1
    for _ in range(n):
        sequence.append(a)
        a, b = b, a + b
    return sequence
num_terms = 10
print("Fibonacci Sequence:", fibonacci(num_terms))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task8.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task8.py"
Fibonacci Sequence: [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 9 :

task9.py > ...

```python
#task 9
#Twin Primes Checker
def is_prime(num):
    """
    Check if a number is prime.

    Parameters:
    num (int): Number to check

    Returns:
    bool: True if prime, False otherwise
    """
    if num <= 1:
        return False
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            return False
    return True
def twin_primes(n):
    """
    Generate twin primes up to n.
    Parameters:
    n (int): Upper limit for twin primes
    Returns:
    list: List of twin prime pairs up to n
    """
    twins = []
    for i in range(2, n - 1):
        if is_prime(i) and is_prime(i + 2):
            twins.append((i, i + 2))
    return twins
upper_limit = 100
print("Twin Primes up to", upper_limit, ":", twin_primes(upper_limit))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task8.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task8.py"
Fibonacci Sequence: [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task9.py"
Twin Primes up to 100 : [(3, 5), (5, 7), (11, 13), (17, 19), (29, 31), (41, 43), (59, 61), (71, 73)]
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 10 :

task10.py > ...

```python
#task 10
#Refactoring the Chinese Zodiac Program
def get_zodiac(year):
    """
    Return Chinese zodiac sign for a given year.
    """
    zodiac = [
        "Monkey", "Rooster", "Dog", "Pig",
        "Rat", "Ox", "Tiger", "Rabbit",
        "Dragon", "Snake", "Horse", "Goat"
    ]
    return zodiac[year % 12]


year = int(input("Enter a year: "))
print(get_zodiac(year))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task10.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task10.py"
Enter a year:
```

TASK 11 :

task11.py > ...

```python

#task 11
#Refactoring the Harshad (Niven) Number Checker
def is_harshad(number):
    """
    Check if number is a Harshad number.
    """
    if number <= 0:
        return False

    digit_sum = sum(int(d) for d in str(number))
    return number % digit_sum == 0


num = int(input("Enter a number: "))
print(is_harshad(num))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task11.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task11.py"
Enter a number: 3
True
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 12 :

task12.py > ...

```python
!
:oring the Factorial Trailing Zeros Program
ıt_trailing_zeros(n):

ıt the number of trailing zeros in n!.

ıt = 0
:r_of_5 = 5
.e n >= power_of_5:
 count += n // power_of_5
 power_of_5 *= 5
ırn count
: int(input("Enter a number: "))
lumber of trailing zeros in", number, "factorial is:", count_trailing_zeros(number))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task12.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task12.py"
Enter a number: 5
Number of trailing zeros in 5 factorial is: 1
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 13 :

task12.py > ...

```python
!
:oring the Factorial Trailing Zeros Program
ıt_trailing_zeros(n):

ıt the number of trailing zeros in n!.

ıt = 0
ɜr_of_5 = 5
.e n >= power_of_5:
 count += n // power_of_5
 power_of_5 *= 5
ırn count
: int(input("Enter a number: "))
lumber of trailing zeros in", number, "factorial is:", count_trailing_zeros(number))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task12.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task12.py"
Enter a number: 5
Number of trailing zeros in 5 factorial is: 1
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 14 :

task14.py > ...

```python
#task 14
#Lucas Number Sequence – Test Case Design
def count_trailing_zeros(n):
    """
    Count trailing zeros in n! without computing factorial.
    """
    if n < 0:
        return 0

    count = 0
    while n > 0:
        n //= 5
        count += n

    return count


num = int(input("Enter number: "))
print("Trailing zeros:", count_trailing_zeros(num))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task14.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task14.py"
Enter number: 5
Trailing zeros: 1
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```

TASK 15 :

task15.py > ...

```python
#task 15
#Vowel & Consonant Counter – Test Case Design
def count_vowels_consonants(text):
    """Return count of vowels and consonants."""
    vowels = "aeiouAEIOU"
    v_count = 0
    c_count = 0

    for char in text:
        if char.isalpha():
            if char in vowels:
                v_count += 1
            else:
                c_count += 1

    return v_count, c_count


print(count_vowels_consonants("hello"))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop/ai coding/10.3.py/task15.py"
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py % /opt/homebrew/bin/python3 "/Users/bandarignaneshwar/Desktop
/ai coding/10.3.py/task15.py"
(2, 3)
bandarignaneshwar@BANDARIs-MacBook-Air 10.3.py %
```